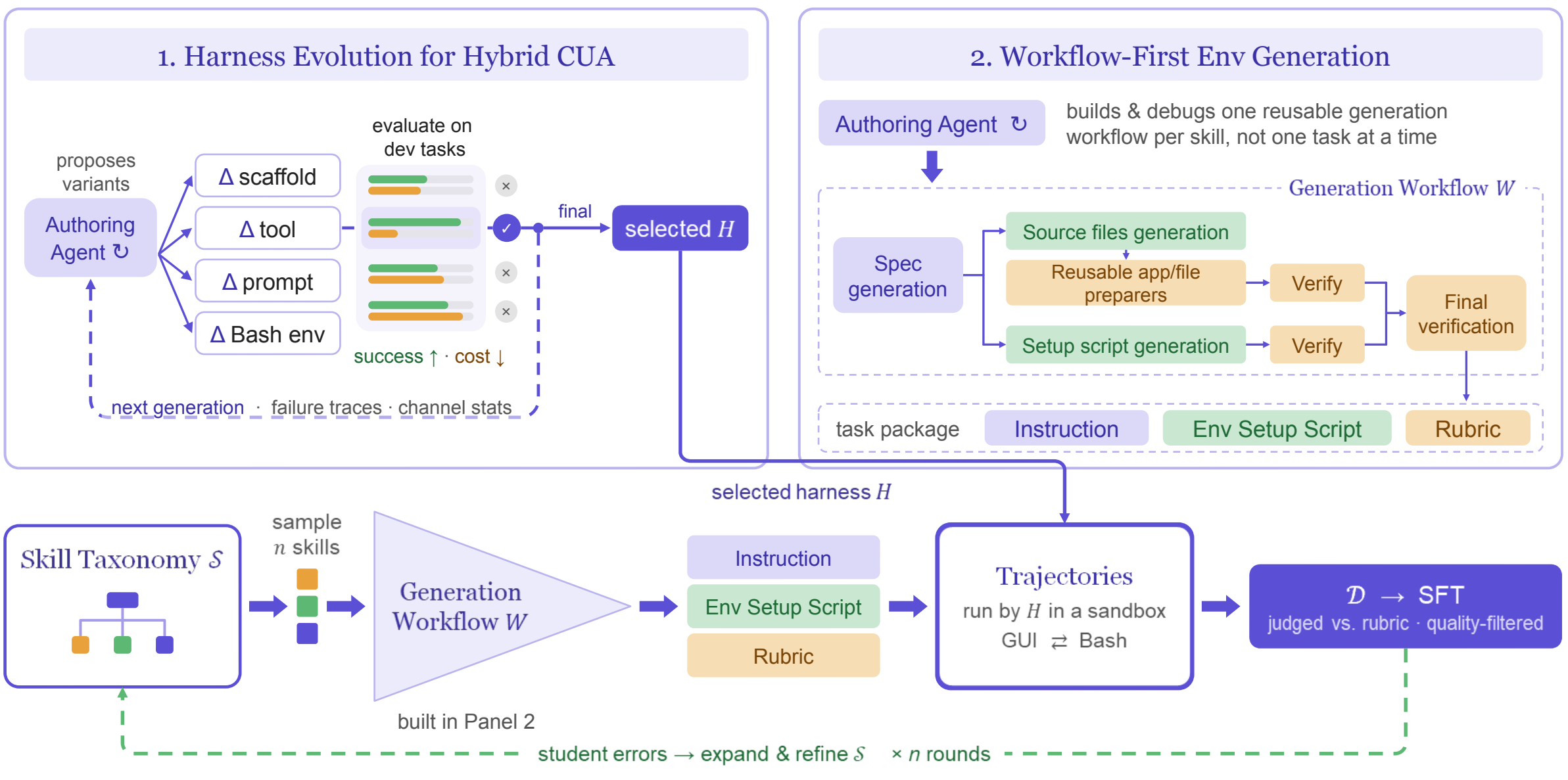

1. Harness Evolution for Hybrid CUA
proposes variants
Authoring Agent ↻
Δ scaffold
Δ tool
Δ prompt
Δ Bash env
evaluate on dev tasks
success ↑ · cost ↓
final
selected H
next generation · failure traces · channel stats
2. Workflow-First Env Generation
Authoring Agent ↻
builds & debugs one reusable generation workflow per skill, not one task at a time
Generation Workflow W
Spec generation
Source files generation
Reusable app/file preparers
Setup script generation
Verify
Verify
Final verification
task package
Instruction
Env Setup Script
Rubric
selected harness H
Skill Taxonomy S
sample n skills
Generation Workflow W
built in Panel 2
Instruction
Env Setup Script
Rubric
Trajectories
run by H in a sandbox
GUI ⇄ Bash
D → SFT
judged vs. rubric · quality-filtered
student errors → expand & refine S × n rounds